

Week - 2 [DFS]

CODE :

```
def dfs(initial, goal):
    stack = [(initial, [])]
    visited = set()

    while stack:
        board, path = stack.pop()

        if board == goal:
            return path

        if board in visited:
            continue
        visited.add(board)

        b = list(board)
        idx = b.index(0)
        r, c = idx // 3, idx % 3

        for dr, dc, move in [(-1,0,'U'), (1,0,'D'), (0,-1,'L'), (0,1,'R')]:
            nr, nc = r + dr, c + dc
            if 0 <= nr < 3 and 0 <= nc < 3:
                n_idx = nr * 3 + nc
                b[idx], b[n_idx] = b[n_idx], b[idx]
                nxt = tuple(b)
                if nxt not in visited:
                    stack.append((nxt, path + [move]))
                b[idx], b[n_idx] = b[n_idx], b[idx]

    return None

initial = (2, 8, 3, 1, 6, 4, 7, 0, 5)
goal    = (1, 2, 3, 8, 0, 4, 7, 6, 5)

res = dfs(initial, goal)
if res is not None:
    print(f"Initial State: {initial}")
    print(f"Final State: {goal}")
    print(f"Success! Found in {len(res)} moves.")
```

OUTPUT:

```
Initial State: (2, 8, 3, 1, 6, 4, 7, 0, 5)
Final State: (1, 2, 3, 8, 0, 4, 7, 6, 5)
Success! Found in 13441 moves.
```